

THE FEDERAL POLYTECHNIC, MUBI

DEPARTMENT OF COMPUTER SCIENCE

LECTURE NOTE

ND II COMPUTER SCIENCE

MANAGEMENT INFORMATION SYSTEM (COM 224)

BASIC COURSE OUTLINE

1. Understand the concept of system and systems theory.
2. Understand the concept of management information.
3. The features of management information systems (MIS)
 - a. Transaction processing.
 - b. Office automation.
4. The different applications of MIS.
5. The principles of decision making
6. The development cycle of an MIS
7. The principles of project management.

CONCEPT OF SYSTEM THEORY

System

A system can be described as a group of interrelated elements that are integrated with the common purpose of achieving a particular objective by accepting inputs and producing outputs in an organized transformation process. By this definition, an organization such as a firm or a functional area is a system. The organization consists of the physical resources like; Manpower, Material, Machines, Money, and conceptual Information (including data) and they work towards achieving particular objectives that are specified by the owners or management. Other examples of system are biological system, business system, electrical system, data system, education system, operating system, mathematical system, sound system, waste disposal system, computer system, queuing system, etc.

Subsystem

A Subsystem is simply a system within a system . This means that systems exist on more than one level that is, Systems can be composed of subsystems or elemental parts. For example, the system of the firm includes subsystems of the bank may be such departments as savings, demand deposit (checking accounts), and installment loan.

Super system

When a system is part of a larger system, the larger system is the super system. For example, the system of the firm exist in one or more larger environmental systems or super systems. If the firm is a bank, for example, it is part of the financial community. It is also part of the business community, the local community, and the global community.

The manager's major responsibility is to ensure that the firm meets its objectives, by ensuring that the various. parts of the firm work together as they should. These definitions are illustrated in the following diagram:

Firm being a subsystem within super systems and systems:

Basic Elements of System

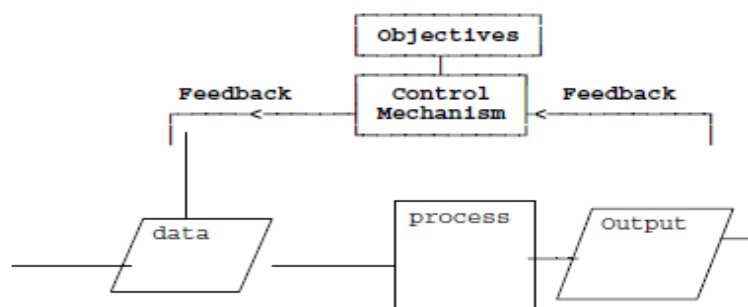
A basic configuration of system consists of following elements;

Input element, Transformation process, and Output element. Additionally, some systems have feedback loop, Control mechanism, and Objectives.

Input resources are transformed into output resources. The resources flow from the input element, through the transformation element , and to the output element . A control mechanism monitors the transformation process to ensure that the system meets its objectives. The control mechanism is connected to the resource flow by means of a feedback loop, which obtains information from the system output and makes it available to the control mechanism.

The control mechanism compares the feedback signals to the objectives and directs signals to the input element when it is necessary to change the system operation.

Diagrammatic relationship of system Elements



A feedback is data about performance of a system.

CLASSIFICATION OF SYSTEMS

There are three categories of systems; Probabilistic system, cybernetic system and deterministic system.

1. Probabilistic system and Deterministic System

Probabilistic system is also called stochastic system is the one whose outcome cannot be predicted with precision. That is, not certain what outputs will be achieved from specific input because these systems are subjected to random influences from the internal and external environment. Therefore, it cannot be predicted precisely. Examples include the business system, economic system and particularly agricultural system.

2. Deterministic System

A deterministic system also called mechanistic system is one whose behavioral patterns can be predicted if its present state and operation characteristics are known. Such a system operates according to a predetermined set of rules. A good example of a deterministic system is a computer program. The computer programmer knows in advance the output his program will produce, if the program is error free.

3. Cybernetic System

A cybernetic System is one that automatically adapts to its environments. A system with feedback and control components is sometimes called a cybernetic system, that is, a self-monitoring, self-regulating system. The behavior of the cybernetic system is a subject of rigorous scientific investigation.

Adaptive Systems: These are dynamic system which responds to changing circumstances by adjusting its behavior on a self-organizing basis.

The system alters its parameters as a result of measuring output. Example includes organizational systems such as human beings. The degree to which a business is a adaptive depends to a great extent on the caliber of its management.

SYSTEM THEORY

Open System

This is a system that interacts with other systems by exchanging input and output components with its environmental elements such as customers, suppliers, competitors, the labour force, government and global community. A firm and most systems that occur naturally in business are examples of open system. It has to adapt to its changing environments where necessary in order to ensure its continued existence. Also, a business system must be capable of reorganization in order to cope with the changing market situations.

Closed System

This is a system whose function and behavior is not affected by its environment. That is, it is self-contained and the input/output elements do not interface with environmental elements outside it. Example includes chemical reactions in a controlled laboratory.

Nature of a system

There are essentially four (4) views that support the General System Concept. These include the following attributes:

- a.) Synergy: All systems are made up of subsystems and for a meaningful Management Information System, the subsystems must act as one body where the whole is the greater than sum of all the subsystems.
- b.) Hierarchy: The system when viewed as a whole, is made up of subsystems in which control flows in an ordered sequence from the top to the bottom.
- c.) Interdependency: All the parts of a system constitute an indissoluble whole so that no part can be altered without affecting the other parts.
- d.) Sub-optimality: This is a situation in which the sub-systems become selfish and selfcentered and pursuing their own objectives independently. The overall effect may be detrimental to the existence of the whole organization.

CONCEPT OF MANAGEMENT INFORMATION

Management

Management can be simply defined as the organization and mobilization of all human and material resources in a particular system for the achievement of identified objectives in the system. The two key words are '*human*' and '*materials*'. It is the business of management that both human and material resources within a system should be organized and mobilized so that the objectives of the system can be achieved to the fullest. Then administration or management therefore influences the results to be achieved, the direction to be pursued and the priorities to be recognised.

Management functions

These can be applied to organization, be it industry, education or government. These elements are:

P=Planning

O=Organising

D=Directing

C=Co-coordinating

C=Controlling

- **Planning:** This is the first step in any administrative function. It involves making arrangements for the future. It deals on how to use time, personnel and equipment in future. The future may begin with days, weeks, month or year. Plans are made after or goals and objectives are set. Planning is usually regarded as a search for the best method of achieving these goals.
- **Organising:** The manager decides What jobs would have to be filled and the duties and responsibilities attached to each one. Effective organization implies that the right people are assigned to the correct jobs at the appropriate time. Activities should be assigned to individuals on the basis of their ascertained qualification, experience and interests. On the part of the manager, a lot of wisdom, maturity and objectivity are required.
- **Directing:** Directing involves giving staff details on how to go about their jobs. It is not once for – all activity but an on-going one. Officers have to be continually put on the right path.
- **Co-ordinating:** This relates to the activities of different units to one another to ensure that the different activities are met, to achieve the set objectives. Coordination occurs when the organisation's numerous activities are planned, organized, directed and controlled.
- **Controlling:** It is the measuring and correcting of activities are contributing to the achievement of planned goals.

Some authors have pointed out that appraisal, evaluating and supervision are the administrator's tools for controlling, directing and leading.

CATEGORIES OF INFORMATION

There are three main categories of information and these are related to the purpose for which the information is utilized. The three categories of information are: Strategic, Tactical and Operational information.

i) Strategic Information (top management): This is used by senior managers to plan the objectives of their organization, and to assess whether the objectives are being met in practice. Example is Board of Directors whose duties include to plan product range, plan expansion into new markets and to establish at least 5-year plan and annual plans for the corporation. Information needed includes highly summarized

information regarding corporation activities to-date e.g sales, transactions, Forecasts, awareness of competitor activities, and awareness of overall economic climate.

ii) Tactical Information (middle management): This is used by middle management to decide how the resources of the business should be employed and to monitor how they are being and have been employed. Example, Sales Managers whose duties include ensuring that sales target are met, determining monthly sales plan, and correcting variances from plan. Examples of information needed -monthly sales analysis, market conditions, competitors' activity.

iii) Operational Information (low-level management): This is used at the lower-level management for daily or hourly running of their section. Users of operational information includes foremen, sectional heads, sales clerk etc, to ensure that specific tasks are planned and carried out properly within the factory or office. Information needed includes prices of items, customers credit rating, stock availability etc.

Characteristics of Operational Information

- i) derived almost entirely from internal sources
- ii) is highly detailed
- iii) relates to the immediate term
- iv) is task-specific
- v) is prepared constantly or very frequently
- vi) is largely quantitative

Characteristics of Tactical Information

- i) it is relevant to short and medium term
- ii) describes or analyses activities or departments
- iii) it is prepared routinely and regularly
- iv) it is based on quantitative measures (cash flow forecasts, budgetary control or variance analysis reports, short-term purchasing requirement).

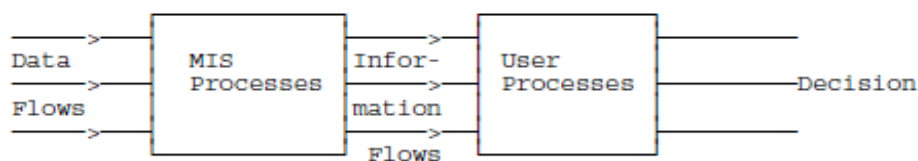
Characteristics of Strategic Information

- i) derived from both internal and external sources
- ii) summarized
- iii) relevant to the long term
- iv) deals with the whole organisation
- v) it is both quantitative and qualitative (eg future market prospects, total cash needs capital equipment needs).

MANAGEMENT INFORMATION SYSTEM (MIS)

A Management Information System (MIS) can be defined as an assemblage of data (e.g. facts and opinions) that are processed (e.g summarized, categorized, or projected) in such a way that they result in intelligent information that management can use to make decisions, and that the organization can use to attain its goals.

MIS can also be defined as the combination of human and computer based resources that results in the collection, storage, retrieval, communication and use of data for the purpose of efficient management of operations and for the business planning.



The main objectives of MIS is to use formalized procedures to provide management with appropriate information from all relevant sources, which would enable the manager make timely and effective decisions. The information must be provided at an acceptable level of accuracy and at an economical cost.

Organizational information systems could include the following:

- i. Marketing Information System,
- ii. Manufacturing Information System,
- iii. Financial Information System,
- iv. Human Resources Information System,
- v. Information Resources Information System.
- vi. Executive Information System.

STRUCTURE OF MANAGEMENT INFORMATION SYSTEM (MIS)

1. Strategic Planning and control: “strategic planning is the process of deciding on objectives of the organization, on changes in these objectives, on the resources used to attain these objectives, and on the policies that are to govern the acquisition, use, and disposition of these resources.

2. Management control: This is the process by which managers assures that resources are obtained and used effectively and efficiently in the accomplishment of the organization’s objectives.

3. Operational control and Planning: This means “the process of assuring that specific tasks are carried out effectively and efficiently.”

The basic difference between management control and operational control is that operational control is concerned with tasks (such as manufacturing a specific part) whereas management control is most often concerned with people.

The importance of MIS to business organizations

The importance derived from implementing a management information system includes the following:

1. Provision of improved services to customers.
2. It facilitates decision making process of managers.
3. The routine processing of business transactions is faster and cheaper than manual method.
4. It fastens the planning and control functions of management.
5. The information contents of MIS can be used by many managers for their varied needs.
6. It can be used to achieve competitive advantage in the marketplace.
7. Expected cost savings/reductions
8. Increase in output volumes and Increase in profit
9. Increase in accuracy level
10. Improvement in working environment
11. Improvement in turnaround time
12. Tangible benefits, e.g customer relations and achievement of set goals.

Characteristic features of MIS

- a) MIS supports structured decision at the operational and management control levels. However, they are also useful for planning purposes of senior management staff.
- b) MIS are generally reporting and control oriented. They are designed to report on existing operations and therefore to help to provide day-to-day control of operations.
- c) MIS rely on existing corporate data and data flows.
- d) MIS have little analytical capability.
- e) MIS generally used in decision making using past and present data.
- f) MIS is relatively inflexible.
- g) MIS has an internal rather than an external orientation.

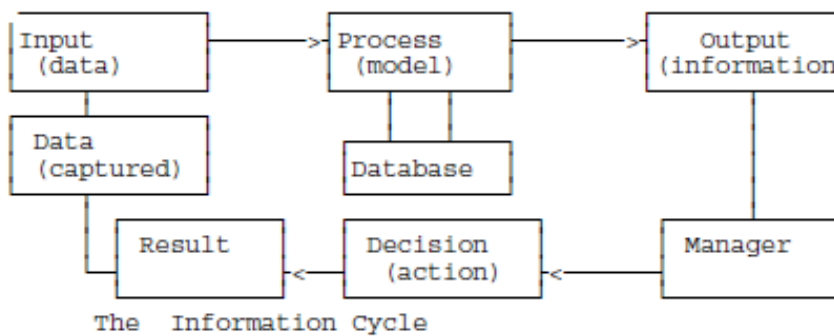
THE CONCEPT OF TRANSACTION PROCESSING

The terms data and information are often used synonymously. To computer professionals they are different. **Data** is a piece or collection of raw facts representing people, objects, and the day-to-day activities of an enterprise. An enterprise in this case can be a hospital, a school, a government ministry, a business organisation, etc. *Data is a valuable resource in any organisation.*

Information is data that have been put into a meaningful and useful context and communicated to a recipient who uses it to make decisions. Information involves the communication and reception of intelligence or knowledge.

Information consists of data, Images, text, documents, and voice, often inextricably intertwined, but always organized in a meaningful context. Information does not always travels directly from the physical system to the manager. Many managers are located some distance from the physical activity. These managers must obtain information from a system or procedure that produces the information from gathered data.

A simple schematic that represents how we will think of information is shown in following figure.



QUALITIES/ATTRIBUTES OF INFORMATION

- 1)Accessibility: This refers to the ease and speed with which information can be obtained.
- 2)Comprehensiveness: This refers to the completeness of the information. Incomplete information affects management decisions that is, it leads to incomplete management action.
- 3)Accuracy: This pertains to the degree of freedom from error of the information, especially when large volumes of data are involved. Transcription, transposition and computation error are examples.
- 4)Appropriateness: This refers to how well the information relates to user's request. This information must be relevant to the matter at hand.
- 5)Timeliness: This relates to the elapsed time of the cycle ie input, processing and reporting of output to users. For the timeliness of information to increase, the duration of this cycle must be reduced.
- 6)Clarity: This refers to the degree to which information is free from ambiguity.
- 7)Flexibility: This refers to the adaptability for use by more than one user.

DATA PROCESSING STAGES

Data Processing (DP): Activities applied in the transformation of data into useful information. Techniques for collecting, assigning meaning to, communicating, and retaining data for later use.

Data Processing activities (Producing Information From Data)

Data processing consists of gathering the raw data input, evaluating and bringing order to it, and placing it in proper perspective so that useful information will be produced.

All data processing whether done by hand or by the latest computer system, consists of three basic activities: capturing the input data, manipulating the data, and managing the output results.

1. Capturing: Data must be originated in some form and verified for accuracy prior to further processing. They may initially be recorded on paper source documents and then converted into a machine usable form for processing, or they may be captured directly in a paperless machine-readable form.

2. Verifying: Refers to the checking or validating of data to ensure that it was captured and recorded correctly. Examples are ; a person reviewing another's work, the use of check digits in coding structures e.t.c.

3. Manipulating the data. One or more of the following operations may then have to be performed on the gathered data:

- i. **Classifying:** It places data element into specific categories which provide meaning for the user e.g Sales data can be classified as inventory type, size, customer, sales-woman, warehouse shipped from.
- ii. **Arranging (sorting):** Places data element in a specified or predetermined sequence, e.g An inventory file can be arranged by product code, activity level, Naira value
- iii. **Summarising:** Combines or aggregates data element, e.g reducing data in the logical sense, as in the example where personnel manager wants a list of names of employees assigned to a department in a form.
- iv. **Calculating:** Entails the arithmetic and/or logical manipulation of data e.g computation to derive employees pay, customer's bill, students grade point averages. Sophisticated calculations include linear programming, fore-casting e.t.c.

4. Managing the output results. Once data have been captured and manipulated, one or more of the following operations may be needed:

- i. **Storing:** Places data onto some storage media such as magnetic tape, disk e.t.c whers it can be retrived when needed.
- ii. **Retrieving:** Entails searching out and gaining access to specific data elements from the medium where it is stored.
- iii. **Communicating.** Transferring data from one location or operation to another for use or for further processing is data communication- a process that continues until information, in a usable form, reaches the final user.
- iv. **Reproducing:** Duplicates data from one medium to another or into another position in the same medium e.g. a file of data stored on a magnetic disk may be reproduced onto another magnetic disk or onto a magnetic tape for further processing or for security reasons.

METHODS OF INFORMATION/DATA PROCESSING

1. Manual Method: Involves operations performed by a clerk assisted, if defined, by specific aids such as a pocket calculator or adding machine etc. This method of processing is only suitable in cases where there is no emphasis on the specific period of time the work must be completed. The processing is simple and, in most cases,, the employment of more hands will solve the problem of time constraint. Example is the updating of handwritten ledger records.

2. Mechanical Method: Involves operations that are performed by machine consisting of a keyboard but which are actuated by depressing appropriate keys by an operator. Example is the posting of lodgements and withdrawals on the personal customer's ledger using an accounting machine.

3. Electronic Method: Is more or less like the mechanical method except that in this case the machine being used is electronic and may have optional peripheral devices that are attached to it. It can be used for many accounting procedures including payroll, stock schedules and ledger updating. These days computers are being used. Data for input are entered through keyboard, card reader, optical character/mark reader, tape reader etc, while processed data are printed or displayed or stored on auxiliary storage. Its processing speed is high and can therefore handle quite a large amount of data. Very suitable for processing tasks that are repetitive in nature.

Advantages of Manual system

1. Initial cost of setting up is cheaper.
2. Major training is not needed.
3. Economical for small business.

Advantages of computerized system

1. Speedy operation.
2. Automatic operation.
3. Accurate information.
4. Varied management information
5. Random enquiries on stored data can be easily performed.
6. High productivity and improved service.

THE CONCEPT OF OFFICE AUTOMATION

Office Automation System or office information systems

Office automation refers to a wide variety of computer-based and non-computer-based applications that make office workers more productive at their jobs. Office automation includes all of the formal and informal electronic systems primarily concerned with the communication of information to and from persons both inside and outside the firm. Office automation applications have grown one by one until the extent that exist today.

These are:

-Electronic Mail, -Voice mail, -Audio conferencing, - Video conferencing, - Facsimile transmission (FAX)
-Desktop publishing

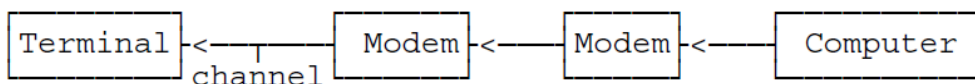
Many organisations and many managers will only embrace what the organisation needs and can afford when incorporating OA.

DATA COMMUNICATION, COMPUTER NETWORKS/INTERNET

DATA COMMUNICATION

Data communication is the movement of coded data and information from one point to another by means of electrical, or electromagnetic devices, fibre-optic cables, or microwave signals. It is equally called teleprocessing, telecommunications. In data communication, there must be the sender, receiver, of communication as the main components of the system. Many technologies are used in the channels, but the most common is the telephone line.

Data communication Schematic



A **network** is a group of interconnected computer systems sharing services and interacting by means of a share communications link. In a peer-to-peer environment, it may be described as a group of computers connected to each other (in some cases, with a server) such that can share resources such as document, files and / or printers.

Networking is a way of connecting more than one computer under the same or different operating environment, at different location possibly far distances, using the same or different medium of information exchange and protocols, with the aim of sharing information, data, and other resources.

By this illustration computer communications have to do with the transfer of information by:

- i.) Direct connection by cable of two computers.
- ii.) Connection of two computers via the public telephone system or other telecommunications links.
- iii.) Networking of a number of computers.

"**Telecommunications**" means 'communications at a distance'. Telecommunications technology embraces radio waves travelling through the air or through space, electrical waves flowing along a telephone wire, and laser pulses travelling along optical fibres.

Reasons for computer Communications

There are a number of reasons why you might want to connect your computer with others:

- i.) To transfer files between your computer and a different type of computer with incompatible disks. For example, you might wish to transfer a document written on a PC to an Apple Mac running a DPT system.
- ii.) To send messages to people with other computers, and to receive messages from them. This is called electronic mail or E-mail for short.
- iii.) To join user groups or mailing lists, which is a kind of on- line bulletin board.
- iv.) To access information stored on other remote computers, such as sales, inventory, and other internal information in the case of your organization's computers, or financial and economic information in the case of public system.
- v.) To update your organization's computer files when you are away from base.
- vi.) To share files with other users in your office or organization.

Hardware and software requirements for computer communications

- i.) To get your computer communicating with another, you need communications Software and a suitable cable.
- ii.) If you wish to carry out these communications via the telephone network, you need a modem.
- iii.) To network your computer to others in your office, you need a local area network.

TRANSMISSION METHODS

Transmission modes determine the one-way or two-way traffic flow between devices. There are three methods of transmission, these are listed below:

- i.) Simplex, ii.) Half-duplex and, iii.) Full-duplex method.

i.) Simplex moves data in one direction only. One device sends and one device receives just like radio and television communication.

ii.) Half-duplex in contrast, allows two-way communication, but only one way at a time. devices take turns sending or receiving, similar to CB radio communication.

iii.) Full-duplex :To communicate in both directions simultaneously, we use full-duplex mode. Separate transmission signals prevent simultaneous communication from interfering with each other.

Conceptually, a full-duplex line is equivalent to two simplex lines, one in each direction.

PROTOCOLS

communication requires a strict set of rules or procedures, called protocols for maintaining communication. Protocols define the technical details of communication, such as rules for initiating conversations, steps followed for sending and receiving, control characters, dialog procedures, and error detection techniques. Protocols are sometimes built into the hardware by the manufacturer. If two devices do not use the same protocol, they cannot communicate with each other.

NETWORK TOPOLOGIES

The word topology implies a description of how various elements within a system are related physically to each other. In the context of networking systems, we use several different ways of describing how computers might connect to each other via a cabling system. These different ways of connection relate to the different types of network protocol involved. There are three main types of networking topology, these being the Bus, Star and Ring topologies.

Bus Topology

The bus topology describes system where each computer is linked to another by tapping into a common cable. The network system then shares a common single cable which runs from one computer to another. The bus topology has one distinct advantage over other cabling systems.

Information can be transmitted from one computer directly to another without travelling through every other nodes. The failure of one network computer does not affect the other network computers.

It uses a single cable and does not require additional hardware to implement the network. This makes this system relatively easy and inexpensive to implement.

Star Topology

The star topology describes a system where all computers are connected to a central distribution system (i.e. a host computer). The host can be a mainframe or a smaller computer. Usually, the database and the printer are part of this host computer. The other nodes are attached to the host, and all messages are routed through the host. If the central computer fails, so does the network. This system allows for intelligent control systems to be incorporated into the distribution hub.

Ring Topology

Ring topology is the physical layout of a local network in which all nodes are attached in a circle, without a central host computer. Messages travel around the ring until they reach the computer to which they are addressed. If the ring is broken, the network fails.

INTERNET

The Internet is a global network of participant connected by computers. Massive amounts of information travels through millions of channels around the world forming the Net. The word Internet literally means network of networks. In itself, the Internet comprised thousands of small regional networks scattered throughout the globe.

Many people may tell you, the "Internet is not owned by anyone". However, this is not entirely correct. Different commercial companies let you connect to the Internet and access the information stored on the Internet. They are called Internet Service Provider (ISP). Just as the Phone Company or Nitel provide you with a telephone line to contact anyone in the world, an ISP provides you with a computer line to connect to anyone in the Internet world. There is no organization responsible for the information stored on the Internet. The suppliers themselves are responsible for the information they provide. No one controls the information; the information is free and you can get as much as you want in expense of your telephone bills and your credit points with your ISP.

How does the Internet works?

Corporations, Institutions, and individuals, according to their means put information into the Internet. Information, they say, is power. The Internet will provide you with as much information as you can handle, once you know how to browse the net.

Connecting to the Internet

Connecting to the Internet requires that you have some basic hardware and software in place:

1. A personal Computer (PC)
2. A modem (with a telephone line)
3. An ISP B an ISP provides access to the Internet. It should also, in many cases, provide all necessary software for connecting your computer to the Net. It also provides an Internet tools software kit such as a Web Browser or an e-mail application.
4. An operating system (most commonly used today is Microsoft Windows 7,8/10 or Mac O/S)
5. An Internet Browser, e.g. Microsoft Internet Explorer, Netscape, Eudora, Lynx, Mosaic. etc.

File Transfer Protocol

As the name implies, it facilitates moving files from one computer to another. It has become the common language for sharing data.

How e-mail works?

E-mail works a lot like standard mail, but it is only faster. Internet users call standard mail 'snail mail' because it is so slow. In an e-mail environment, everyone has a mailbox with an address, just like with standard mail. The difference is that an e-mail mailbox is electronic and its stored in an ISP server. Your ISP function in the same way the local post office near our homes does.

APPLICATION NEEDS

MICROCOMPUTER SYSTEMS are used in business to perform a wide variety of tasks, several of which are listed below:

- i.) Transaction Processing. (ii.) Analysis and decision making.
- iii.) Planning, Coordinating, scheduling, and organising. (iv.) Retrieval and update. (v.) Reporting.
- (vi.) Writing. (vii.) Presentation. (viii.) Communications. (ix.) Learning. (x.) Program development.

Types of Software

There are three main types of software. They are:-

1. The system program
2. The application program (package)
3. The user application program.

THE SYSTEM PROGRAM: The system programs are those programs written by the computer manufacturers. These programs are supplied with the computer system on purchase. The system programs includes the operating system, the language translator and the utility or services program/library program.

The Operating system:- The operating system are those system program that interfares between the user of the computer and the computer hardware. The operating system is described as the manager of computer resources. It is the house keeper of the computer resources, and as the housekeeper, it knows the condition and the location of each resources. It allows the computer user to make use of those resources and reclaims them from the user after use the safe keep them again for future use.

Language Translator

The Assembler:- The assembler is the language translator that converts programs written in assembly language to machine language. It translates the assembly programs contained in the source file to the target or object language which the computer can now execute if it is error free Examples are: Microsoft MACRO ASSEMBLER, Microsoft Quick Assembler, and A86 Macro Assembler.

The Compiler:- The compiler is a language translator that converts programs from high level language to machine language. It picks the high level language contained in the source file, converts all to machine language. It create a file called object file by itself and it is doing the translation where it puts the translated version of the source code which is now object code. In essence, the compiler maintains two files i.e *source file and object file*. The object file are in executable form.

The Interpreter:- The interpreter is a language translator that translates the high level language to machine language. It pick the statements line by line and translates them one statements after the other sequentially. Any line translated, is executed by the computer before the next line is translated. If the interpreter meets a syntax error in any line, it will signal the error and the line number. That syntax error must be debugged before the interpreter can go to the next line.

The Utility programs:- The utility programs just like the name sounds is the type of system programs that are utilized by the users of computer to solve one data processing operation or the other. They remain resident in the storage media where they are stored until they are invoiced through a command to perform. Examples of these programs are: copy, format storage, sorting programs, disc cleaning program etc.

THE USER APPLICATION PROGRAM

These are programs written by the computer user as it will suite their own environment. These programs are designed and program to solve the specific problem by a computer specialist who is employed in the company. This type of program is not solved by vendors like the package, rather they are indigenous to an individual or company environment unlike the packages, they are not ready made.

Examples include the following:

- Payroll, - Stock control, - Sales order processing and sales ledger, - Purchase order processing and purchase ledger.

Those that are used for a specialized form of business eg.

- Transaction processing banks, - Airline seat reservation, - Theatre seat reservation, -Hotel room reservation,
- Point of sale software.

APPLICATION PACKAGES

This is a specially designed program written by the professional programmers for solving specific data processing needs, either commercial or scientific. A package is a complete suite of ready made programs with its documentation covering a processing routine. It is usually supplied by a software house or manufacturer. Examples of such applications include: Payrolls, auditing, Stock control, Network Analysis, Dbase, Spreadsheet, Wordperfect, etc.

1. Word processing programs: These are programs that, more or less, turn a computer system into a powerful typewriting tool and more. It helps us to produce letters, reports, magazines, memos etc.

Example of word processing programs are: Microsoft word, Wordstar, Wordperfect. PC Write, Multimate, Ten Best and Wordstar 2000.

2. Database Management: Databases are the backbone of business computing. They store and retrieve information such as customer lists, inventories and notes. They can also sort information, letting users do things like search their customer list for customers in a given STATE.

These programs help us to record and maintain information about people, places, things and management reports of all types. Examples are Dbase III & IV, Foxbase, Oracle, Foxpro and Clipper, MICROSOFT ACCESS, Paradox, Professional File, WindowsFile and Q&A etc.

3. Accounting programs:- Accounting programs keep track of business income and expenditures including accounts receivable, accounts payable, operating expenses, cash flow and payroll.

Popular accounting programs include QuickBooks, DacEasy Accounting, Peachtree Accounting, Profit, One-Write, Pacioli 2000, and the ACCPAC series of programs.

4. Spreadsheet- Spreadsheets are software packages that, more or less, turn a computer system into a sophisticated electronic calculator. Many spreadsheet packages also have presentation graphics generators, which take data and painlessly convert them into bar charts, line charts, and the like, for management presentations at meetings. These programs help us to produce financial reports and reports. Examples are Lotus 1-2-3, Quatopro, SuperCalc, Multiplan, VP Planner, Planning Assistant and Microsoft Excel.

5. Graphics programs. Graphics programs create graphs and drawings that can be used in newsletters, posters, advertisements and other documents. Some allow users to import photographs into documents or create animated pictures for use in "multimedia" presentations.

Popular graphics programs include Corel Draw, Print Master, Print Shop, Microsoft PowerPoint, etc

6. Desktop publishing. Desktop publishing programs are extremely sophisticated word processors that can incorporate word processing and graphics files to create newsletters, manuals, files, advertisements, magazine and newspapers. Popular DTPs include Microsoft Express Publisher, Aldus PageMaker, Ventura Publisher and Quark XPress.

7. Integrated Packages/All In One. Integrated programs are a program that combines the ability to do several general-purpose applications including three or four different functions, usually word processing, database, graphics and spreadsheet in one program. The most popular integrated packages include Microsoft office, WordPerfect Works, ClarisWork, GeoWorks Pro and Lotus Symphony.

8. Computer aided design. Computer Aided Design is a package that can be used to produce drawings of professional quality on paper as large as your printer or plotter can handle. Such drawings like engineering plans, models, maps, electrical circuit diagrams, planning diagrams, architectural and more are made possible by the use of computer. CAD is widely used in Industry, Commerce, and Education. Popular CAD Packages which can perform all these functions include like Autocad, Auto-Sketch etc.

Guideline for Acquisition of Application Software

1. The software to be Acquired must be one whose use will improve the particular operating function of that business.
2. It must be of high performance.
3. The software to be acquired must be one with enough ease in learning.
4. The software to be acquired must be available when needed.
5. If a software package is complex then training is required.
6. Good documentation: - The package to be used must be properly Documented. This should include how the package is to be installed, how to use the package and other necessary technical details.
7. Users friendliness:- The package must be easy to use even by non computer specialists
8. Compatibility of the package:-Before a package program can be adopted, it must be compatible with the user system requirements as well as with the users hardware.

Modes of Acquisition of Application Software

- I.) Purchase, (ii.) In-house development, (iii.) Contracting out for custom software.

Advantages of Custom Application Software

1. The needs of the organization are precisely met.
2. The method of inputting/outputting data may be designed to suit users taste.
3. They provide facilities for developing areas where Automation is needed in place of manual method.
4. They are very efficient in their use of system resources. It may require less memory and less disk space.
5. Program may execute faster.
6. Programs can be changed to meet changing conditions.

Disadvantages

1. Development time may be lengthy.
2. Human resources needed for development may not be available
3. Since the organization for which it is developed may be the only user, cost of development may be very high.
4. Cost of maintenance/ upgrading may be high.

Factors to Consider When Selecting Application Packages

- 1) The operating system requirement of the package.
- 2) The hardware requirement of the package.
- 3) Whether the package satisfies users requirement
- 4) User friendliness of the package.
- 5) The cost of the package.
- 6) The integration of the package with other packages.
- 7) The reliability of the package.
- 8) The flexibility of the package.

THE DIFFERENT APPLICATIONS MANAGEMENT INFORMATION SYSTEM

Various Types Information Systems

In early day of business computing, computers were usually used to replace the existing manual systems. But , in this modern day ,managers make decisions to solve problems, and information is used in making the decisions . Information is presented in both oral and written forms by an information processor. The computer portion of the information processor contains each of the following computer-based application areas:

- i.) Transaction processing Systems otherwise known as or Accounting Information System, Electronic Data Processing System.
- (ii.) Information Reporting Systems
- (iii.) Decision Support Systems
- (iv) Office Automation System or office information systems or virtual office. This has been discussed in details in the previous week.
- (v) Executive information systems
- (vi) Artificial Intelligence or Expert Systems.

1.) Transaction processing systems (TPS)

This is a system that processes data arising from the occurrence of a firm's business transactions such as Accounts receivable, Payroll, Inventory Control, Hotel reservation system, Sales order entry, Accounts payable and others. Transaction processing systems are computerized systems that perform and record the daily transactions necessary to the conduct of business. They serve the operational level of business organization hierarchy. These systems include both the accurate recording of transactions as well as the control procedures used in issuing such documents as invoices, customer statements etc.

TPSs are traditionally carried out manually, while are Computerized in many organisations today.

2. Information Reporting Systems (IRS) is an information system that provides predefined types of information to management for relatively structured types of decisions. As opposed to focusing purely on data and efficiency which characterize Transaction Processing Systems, IRS focus on information and, occasionally on effectiveness.

Reports are the mainstay of IRS. The main output of the IRS has been hard-copy reports, it equally produces display output on screens either in continuous scrolled form. These include summary and exception reports that were distributed by the MIS

3. Decision Support Systems: A decision support system is a form of information system that provides tools that help managers make decisions with both internal and information in the manner that best suits the decision

they are currently trying to make. The DSS provides the manager with computing and communications capabilities to develop his or her own decision models and information banks. A DSS is not intended to make decisions for managers, but rather to provide them with a set of capabilities to enable them to generate the information they feel they need to make decisions that are semi structured, unique or rigidly changing and not easily specified in advance unlike TPS.

4.) Executive information systems: The term executive is often used to describe a manager on the strategic planning level.

An executive support systems (ESS) is also known as an executive information system (EIS). It is a decision support system that is designed to meet the special needs of executives. It is a system that provides information to the executive on the overall performance of the firm.

5.) Artificial Intelligence or Expert Systems: Artificial Intelligence is the activity of providing such machines as computers with the ability to display behaviour that would be regarded as intelligent if it were observed in human; that is the ability that can be imparted to computers to enable them display intelligent, humanlike behaviour. Artificial Intelligence has been developed in different areas namely - Expert Systems also known as knowledge- based systems- Natural language - Robotics -Learning etc.

DATABASE MANAGEMENT SYSTEM (DBMS)

The concept of Database

In general, a database system is a computer-based record keeping system i.e. a system whose overall purpose is to record and maintain information and make that information available to user on demand.

Databases are useful at both the managerial and clerical levels in an organization.

At the managerial level are a decision-support tool, for reports can be produced from them which summarise what is going on and give pointers for the future.

At the clerical level they provide an easy way of entering and retrieving data, and dealing with inquiries.

Typical business database applications include:

(i.) Stock records (ii.) Personnel records. (iii.) Customer records (iv.) Accounts, and (v.) Mailing lists.

Classification of Database

Database are classified according to the approaches taken to database organisation. The classes are:-

(a) Relational (b) Network (c) Hierarchical

1. Relational database: is a sequential representation of the files inform of a table having rows and columns where each row represents a record and the columns corresponding to the fields.

2. The hierarchical database: is a tree like representation of the database files. The records of a file are represented in a tree like structure with the fields subordinate to the records in the hierarchy.

3. The Network database: is the representation of the database files. The records relationships are shown by linking.

Features of Database

- 1.Create and maintain (add, delete, and revise records) a database.
- 2.Extract and list all records or only those records that meet certain conditions.
- 3.Make an inquiry.
- 4.Sort record in ascending or descending order by primary, secondary and tertiary fields.
- 5.Generated formatted reports with sub-totals and total.

Managing a database

Managing a data base primarily involves the following six tasks:

- 1.ADD new data to the database
- 2.SORT the database into some meaningful order
- 3.SEARCH the database for types of information
- 4.PRINT data from our database onto formatted reports
- 5.EDIT data on the database
- 6.DELETE data from the database.

Advantages of database

- (1) Reduces the amount of data duplication which occurs using conventional file structures.
- (2) Provides fast and flexible access to information.
- (3) Improves consistency of data stored.
- (4) Provides both data and program independence.
- (5) Encourages integration of functional areas in an organization.
- (6) Avoids duplicating input data to update multiple functionally independent files holding the same data.
- (7) Reduces processing for retrospective changes.

Disadvantages of database

- (1) Takes a long time to design and implement.
- (2) Cost more than conventional systems.
- (3) Require high calibre, experienced and specialised personnel.
- (4) Require expensive hardware and software.
- (5) Recovery is more difficult in case of damage.

Components of database system

Database system involves four major components: data, hardware, software, and users.

i.) Data: - A database is a repository for stored data. In general, it is both integrated and shared.

ii.) Hardware: - The hardware consists of the secondary storage volumes-disks, drums etc.-on which the data resides, together with the associated devices, control units, channels, and so forth.

iii.) Software: - Database Management System (DBMS) is a collection of software products that is designed to provide a systematic, integrated and flexible approach to organising and accessing data.

Examples of DBMS include MSAccess, FOXPRO, Dbase IV, ORACLE, SQL Server etc.

USERS OF DBMS

We consider three broad classes of user.

1. The application programmer, responsible for writing application programs that use the database.
2. The end-user, accessing the database from a terminal. An end-user may employ a query language provided as an integral part of the system, or he or she may invoke a user-written application program that accepts commands from the terminal and in turn issue requests to the DBMS on the end-user's behalf.
3. Database Administrator (DBA): A person responsible for maintenance and organization of all data in a firm's database. He must have a sound knowledge of the structure of the database and of the DBMS. He must also be thoroughly conversant with the organization, its systems, and the information needs of the managers.

Responsibilities of DBA

(1) Deciding the information content of the database: It is the DBA's job to decide what information is to be

held in a database

(2) Deciding the storage structure and access strategy: The DBA must also decide how the data is to be represented in the database and must specify the representation by writing the storage structure definition.

(3) Liaising with users: It is the business of the DBA to liaise with users, to ensure that the data they require is available.

(4) Defining a strategy for backing and recovering: Once an enterprise is committed to a database system, it becomes critically dependent on the successful operation of that system.

In the event of damages to any portion of the database caused by human error or failure in the hardware or supporting operating system, it is essential to be able to repair the data concerned with a minimum of delay and with as little effect as possible on the rest of the system.

(5) Monitoring performance and responding to changes in requirements: The DBA is responsible for so organizing the system as to get the performance that is best for the enterprise and from making the appropriate adjustments as requirements change.

COMPUTERIZATION STRATEGIES

Companies Information System can be organised in many ways. However, the basic computerization strategies employed or the most widely used techniques include the following:

- centralised processing, -decentralised processing, -distributed processing, -batch processing,
- remote batch processing, -real-time processing, -time-sharing (multi-access) processing
- integrated processing, -interactive processing, - Multi Programming, - Multiprocessing.

Basic terms:

OFFLINE: means not connected to and controlled by the processor (C.P.U).

OFFLINE DEVICES: are not directly connected to the computer (processor) but are part of a data processing system. Example are data recording machine (for keying holes onto punch cards.), Folders and Inserters, scheduling machines, , etc.

OFFLINE STORAGE DEVICES is a storage that is not directly accessible to the processor (CPU). That is, not directly connected to and controlled by the C.P.U. offline storage are backing or auxilliary storage. E.g. punch card, magnetic tape, magnetic disk, floppy disk, cassette tape, etc. Such storage is outside the processor.

OFFLINE PROCESSING: uses offline devices to get data input ready for processing by the computer. E.g. payroll system, inventory system. Its data entry system is not by the processor.

OFFLINE DATA COMMUNICATION: systems whose communication lines are not connected directly to the CPU.

ONLINE: Means connected to and controlled e.g. Normal mail service by the processor.

ONLINE DEVICES: are directly connected to the processor. E.g. Good reader, visual display unit, magnetic tape drive, magnetic disk drive, MICR reader, OCR reader, voice recognition reader, etc. All hardware or peripherals are online devices.

ON-LINE STORAGE DEVICES a storage that is immediately accessible to the CPU and controlled by it. Such storage is contained in the C.P.U. Example is primary storage or main storage (CORE).

ONLINE PROCESSING: has its data entry system connected to the C.P.U. It uses online devices and allows direct communication with the computer. It is a technique of processing by the computer by means of terminals or V.D.U.s linked directly to the CPU. Online processing offers access to files by terminals and enables updating to be done the transactions. E.g. Real-time, timesharing, distributed and interactive systems.

ONLINE DATA COMMUNICATION: system has its communicating lines connected directly to the C.P.U. E.g. Electronic bulletin board, distributed system, etc.

Techniques of data processing

1. Centralised Processing: Is a technique of processing where all processing functions or activities are done at a single or central location. If a business organisation is widely dispersed with various units, under centralised processing, the processing equipments (computers) will be at the head office or central location. It is often desirable for batch processing application. It can be offline or online.

Advantages of centralised processing:

- Economy on capital expenditure on one CPU instead of several CPUs at the various units of the organisation.
- Less operating costs since there is no duplication of resources at the units.
- Facilitates the standardisation of application.
- Economic use of hardware- single system implementation.
- total number of data preparation staff likely to be feiver.

Disadvantages of centralised processing:

- Distance of other locations will lead to delay in obtaining data and returning output.
- concentration of specialist data processing in one location.
- corrections and interpretation of input are more difficult because of remoteness of point of origin.
- Large investment at risk if system fails to perform satisfactorily.

2. Decentralised Processing: Is a technique where each location or branch has its data processing equipment and staff. Thus, processing activities are allowed to be carried out at each location or branch. Quite distinct from the centre / central office / headquarters. It is desirable for an application in a multi-location organisation that can not be easily linked telecommunicating lines.

Advantages of decentralized processing:

- no delay in obtaining data and returning results.
- corrections and interpretation of input easier.
- one small system can be developed proved in isolation.

Disadvantages of decentralised system:

- Total cost of hardware and system implementation probably greater.
- specialist staff duplicate at each location.

3. Distributed System: Is a special form of decentralised processing that uses data communicating facilities to connect input/output devices positioned at locations remote from the central processor. It is in which processing to some degree, is done at a site independent of the central computer system. Each of its sites has its own processing requirements. Distributed system gives the ability to process data independent of the central computer, as well as communicate data required by the entire organisation to the central computer for organisation-wide planning and control. Distributed system is not centrally controlled as centralised processing. Each of the remote uses has control over his or her own data.

A user in distributed system can access his own computer as well as the database in the head office. the remote processor are interconnected to form a communicating network. The processing time is reduced, this technique is ideal for situations where remote locations can be connected by telecommunicating lines.

Advantages of distributed processing:

- more economical to install.
- can be operated by user staff with very little training.
- Any break down those not affect the company activities.
- Data can be processed at a number of locations independently.
- enhance the security of data/information since the number of people that has access to information can be controlled.

Disadvantages of distributed processing:

- no central control of access to the system because authority is decentralised.
- microcomputers are used in distributed system and not macro-computers, hence lower capabilities.
- problem of ensuring that standards are strictly adhered.

4. Batch Processing: Involves regular processing of large amount of data. In batch processing, data is processed over a specific period of time. At the end of this period, the entire batch is sent to the data processing department for processing. Delay normally occurs between the time a transaction actually takes place and the time it is processed. Batch processing can be online or offline. When it is online, it means data is collected and transmitted through terminal via telecommunication line connected to the C.P.U. The reliability of batch processing is quite high. E.g. payroll, inventory systems etc.

5. Remote Batch Processing: A system that uses input/output devices positioned at locations remote from their CPUs. Remote batch processing uses data communications facilities to connect this remote I/O devices to the computer.

6. Real-Time Processing: Is a system which can through the use of a remote terminal (e.g. V.D.U) access a file on a direct access storage device (DASD) e.g. DISC and with the same terminal update data in that file on DISC and provide an immediate response for decision making. It a form of processing where processing of data input is instantaneous and the output released to effect the functioning of the environment. It involves updating of files on DISC.

Application of real-time include:-

- Airseats reservation.
- Hotel reservation.
- Banking system using data communicating facilities.
- Industrial processing control.
- Manufacturing status of production order.

Disadvantages of real-time processing:-

- Expensive to design.
- Greater time to develop (real-time requires integrated database support and communicating software).

7. Time-Sharing Processing (Multi-Access): Is a technique where one single C.P.U interacts with many independent users (TERMINALS) given each of them individual attention on a time-slice basis. It is concurrent use of a single computer by many users. Each of the users in time sharing system is given a time-slice (a fraction of the processing time) during which CPU attends to a request from that user (terminal). The CPU can partition itself and run several different request at a the same time.

Characteristics of Time-Sharing

- (a) Concurrency:- a number of users at the same time.
- (b)Independence:- each user acts independently, .i.e unaware that there are other users.
- (c) Immediacy:- all request for the C.P.U receive response immediately.
- (d) Spatial unlimited:- unlimited terminals at very remote locations.

Advantages (Time-Sharing)

- C.P.U memory is economically used. It gives room for multiprogramming.
- computer time is economically used.
- No terminal can monopolise the CPU time because there is time-slice for each job.

8. Integrated Processing: system is one in which two or more applications are merge into one single application. That is, several master files can be incorporated into a single file. Output from one or more systems/applications may provide input to another system.

Types of resistance to system changes

There are three (3) type of resistance to system changes, namely:-

- (i) Aggression:** Aggressive behaviour towards the system may come as an attack upon the system to make the system physically in-operative or ineffective. For example, dumping liquids into hardware inputting wrong data deliberately, etc.
- (ii.) Progression:** Projecting resistance to the introduction of an information system. For example, continuously finding faults with system.
- (iii) Avoidance:** Caused by frustration. Users avoid interactions with the system.

Problem posed by resistance to change for system Analyst

- (a) Change of any kind makes people feel uncomfortable as they do not know how they will be able to cope with it. But when change has threatened the user, the employees will be trying to block the change.
- (b) Unless the system analyst gets full cooperation from the employees during system investigation, system analyst gets only a partial or even a distorted view of the old system. If based on these findings, the new system will prove disastrous.
- (c) Something should be done on the demoralized state of the users, it may lead to mass resignation, absenteeism, lateness and poor performance.
- (d) When the new system is installed, they (staff) may feel that it is imposed on them by top management or data processing department. The new system can be used in half-hearted fashion, using every problem that occurs with its operations to indicate that it is unworkable, and should never have been developed. Incorrect data may be accidentally entered into the system to try to crash it. Such behaviour does not happen with every system, but analyst should do whatever he can do to prevent such situation developing.

Remedy to the problems

The followings will give the staff confidence and make them face the changeover.

- a.) alleviate fear by keeping the staff informed all the time about computer development
- b.) fear is caused by ignorance of the computer and what it must be used for. Therefore background education for staff likely to be affected by the computer.
- c.) Organize brief session introducing the idea of computer and later detail training before changeover occurs.
- d.) analyst to develop training materials.
- e.) open the channel of communication between management and staff concerned.
- f.) Inform and involve staff from the early stage of the plans and projects (use notice board, house magazine, etc.)
- g.) discuss with staff association.
- h.) initiate manpower planning where redundancy and job changes are anticipated
- i.) during investigation, this analyst should inspire confidence in the users
- j.) be sensitive to their fears/make attempt to alleviate their fears
- k..) analyst public relation job starts by informing the staff that the new system is in their own interests.
- l.) once the users are convinced that the analyst is there to help them do their job better they will be willing to co- operate with him in his investigation.
- m.) involving experienced managers in the development team. This will make them feel that they have contributed to the design and that the system is theirs rather than being imposed from the above. They will no more complain during implementation when problem occurs, rather they will sort them out.
- n.) essentially user's acceptability is very crucial.

Reason for resentment, Opposition/Resistance/ During System Analysis/Design

Analyst and designers involved in the introduction of computerised system often encounter resentment and opposition from the existing employees.

Reasons why employees react in this manner\way include:-

- a.) fear of redundancy,
- b.) the feeling that the management is criticizing the old system and by implication criticizing their role in it.
- c.) threaten job security, d.) threaten self-esteem of the staff,
- e.) the feeling that they may be incapable of understanding the new system
- f.) disruption caused by lack of understanding on the technology
- h.) afraid that they will be unable to cope with the new methods

- i.) fear that prospects (promotions) may be altered.
- j.) Fear that their position and personal commitment to work may be devalued if it can be done by a machine (computer).
- k.) because the benefits are unknown, it leads to lack of confidence in the computer.
- l.) fear that the job will become more boring
- m.) fear of loss of control , etc.

Steps to reduce Resistance:

These include the following.

- a.) Consider the past history of the company's operating.
- b.) Consider how the previous changes were process.
- c.) discuss the systems and managers implemented affected.
- d.) involving/including existing employees in the implementation team.
- e.) dispel redundancy by defining definite job description for employees after computerization.
- f.) designer/analyst must show respect for existing employees
- g.) Commencing retraining as early as possible.
- h.) explaining the objectives and benefits of the new system to them at the onset.
- i.) Design system that is easy to use
- j.) Design a reliable system.

THE PRINCIPLES OF PROJECT MANAGEMENT

Project Management

A project is a well-defined sequence of events with a beginning and an end, directed toward achieving a clear goal, and conducted by people within such established parameters as time, cost, resources, and quality. A project is different from what you do every day, because a project goal is a specific, nonroutine event. Being nonroutine, a project requires some planning. How much planning you need depends on the complexity of the project. The more complex the project, the more you have to plan.

Planning is necessary to avoid wasting time, resources and effort.

A project is characterized by :

- i.) Must have a definite beginning and a definite ending meaning the a project is time-bound.
- ii.) Non- repetitive.
- iii.) Utilization of scarce resources.
- vi.) Inability to predict with absolute confidence its final outcome.

The Project Life Cycle

Each project moves through a predictable life cycle of four basic phases with each phase calling for different skills from the protect manager. The phases of a project's cycle are:

- Defining the project; -Planning the project; -Implementing the plan; and
- Completing and reviewing the project.

This will be discussed later in this chapter under system development life cyle.

Project Planning and control

The key elements of project management are planning and control. Deciding and specifying what to do is a function of project plan. Making sure that it is done right is the function of project control. Planning is vital if control is required at later stage.

The following are main steps in planning a project:

1. Identify purpose of project
2. Set objectives ,that is, define deliverables
3. Gather information on resources, timescales

4. Construct a plan- list tasks, timescales and dependencies , allocate task
5. Decide on the tools to be used and identify control systems

Computer In Project Management

The complex nature of modern project makes the use of computers inevitable. The areas of applicability include among others: Project formulation; Project feasibility studies; Appraisal Techniques; Work Breakdown Structures; Network Analysis; Costing and Finance; Material Management; Monitoring and Evaluation; Report Writing. Since Project Management involves these three basic phases: Creating a project; Managing a project; and communicating progress, the applicability of computer to such areas enumerated above can be highly appreciated.

WHAT IS MS-PROJECT (Project management software Package)

MS-Protect is Project management software Package that has capabilities that makes it a necessary tool for project Management. MS-project is a Menu driven program i.e. a program that allows you to interact with the computer by enabling you to select from a group of alternatives, with an highlight of the consequences of selecting any of the various alternatives.

Since the essence of project Management is managing information, MS-project's views and features make is easy to enter organize and present project information. MS-project helps plan projects manage and upgrade information and communicate status once a project is underway. For a menu driven package such as MS-project, the use of a 'mouse'- a device for pointing at things on the screen normally used instead of the keyboard-becomes inevitable.

TYPE OF PROJECT MANAGEMENT TOOLS

A project manager has a range of project management tools available to assist with the planning and control of individual systems projects. These tools are designed to improve the effectiveness of the project management process.

The project management tools available include:

1. Work breakdown structure
2. Critical path analysis
3. Gantt chart
4. Resource histogram
5. Budget.

Work breakdown structure is a tool or method used for breaking down work into manageable units, and then allocating those units to members of the project team . The basic idea of creating a (WBS) is to take the work involved in the project and break it down into smaller and smaller parts, until the project consists of series of work packages . These can then be put into the project plan as activities. Having completed the WBS , it will be possible to construct a project network.

Critical path Analysis (CPA) or Network Analysis:

This is a diagram showing the various activities in a project. The aim of the CPA is to identify how those activities link together and to show the critical path or the sequence of activities where delay will result in the overall project.

Understanding Critical Tasks: A critical task is one that, if delayed, would also cause the completion of the project to be delayed. Critical tasks are said to be on the critical path. Changes to tasks not on the critical path will not have an effect on the completion date of the project. With Microsoft Project you can quickly identify the critical path, so you can focus on the tasks that require the closest management. Knowing tasks are critical also helps you assess priorities, effectively assign resources, and determine the effect of changes on the project.

The Critical Path Method (CPM) is a standard Project management technique for determining which tasks are critical. Based on a mathematical model that takes into account the relationships between tasks, their duration, and any constraints regarding the availability of resources, CPM is also used to critical path in a large or complicated project was a significant challenge for the project manager. Today, Microsoft Project brings the proven power of CPM to your project planning being delayed.

Gantt Chart: A Gantt chart , named after the inventor , shows the activities of a project, the same as CPA. However, these activities are presented as a bar chart with the same start and finishing time clearly identified.

Resource Histogram: This is a stacked bar chart showing the number and mix of staff required over the duration of a project.

Budget : The project budget shows for each month of the project, the proposed expenditure . This will be updated as the project progresses with the actual expenditure and the reason for any variances.

Major Feature of Ms-Project

1. Creating a project by entering Tasks in the Gantt Chart

A task is an individual step that must be completed to complete the project.

2. Outlining tasks: Outlining gives the project a hierarchical detail or “subordinate” tasks fit within the broader groupings or “summary tasks”. It can be used for creating summaries, reports, etc.

3. Assigning task relationship by linking tasks in a project by drawing a line connecting tasks in the default Gantt Chart view.

4. Entering resources and assigning them to task in the task entry view .

5. Checking resources usage in the resource pool, as way to manage resources .

6. Resolving resources over allocation ,as way for managing resources.

7. Producing a report: It is possible to carry out the following activities under this heading: change text format, change Gantt Charts, change field displayed in PERT nodes , and print views and reports.

8. Create a template that contains only the features you want in a project file.

9. Work with the critical path and identify critical tasks.

10. Display project information in the new calendar view.

11. Create recurring tasks that are placed at regular intervals in a schedule.

12. Use new filtering options and apply filters with a button.

13. Use the tracking toolbar to get quick access to the commands you need when updating the project.

SYSTEM DEVELOPMENT CYCLE OF MIS

Traditional Systems Development Life Cycle: is a sequence of activities required to develop, implement, and install a new or revised system. That is, an organized way of creating a system whereby the system under study is professionally developed in stages, each of which must be relatively completed before the next stage is began.

If a system is to succeed, all parties must be actively in determining what a system is to do and how. For successful development and implementation of new systems, this organized study of the current system is needed. Standard activity phases are considered bellow.

1. System Objective

2. System Investigation

3. System Analysis

4. System Design and Development

5. Implementation.

6. Review and Maintenance.

1. Objective of a System

Every System is designed to solve some particular problems being encountered by management. For instance, personnel in a functional area may find that an existing system is out of date, a manager might desire certain new information which he requires for better decisions. The moment the problems are articulated, a set of system objectives are developed by the management to satisfy the users. Objectives are usually set to:

- i.) alleviate the worker's workload,
- ii.) Increase in sales and profit,
- iii.) Increase operational efficiency,
- iv.) improve customer service,
- v.) improve employee morale.
- vi.) Satisfy government requirement.

2. Systems Investigation: This is the point at which the proposed new system gains its first exposure and scrutiny. Before a new design and development effort takes place, a problem is identified.

The main purposes of the investigation are to determine:

- If there is a problem, -the extent, or seriousness of the problem, and -Whether to proceed with a full-scale study of the problem and alternative solutions.

3. Systems Analysis

Having investigated all facts about the current system the current system must be analyzed in terms of its problems, working methods, bottlenecks, inefficiencies and information needs of the end user and organizational environment for the purpose of developing a new or improved system / functional requirement of a system that can meet the needs of end users.

The stages of System analysis are:

- i.) Define the problem.
- ii.) plan the project.
- iii.) Collect the facts through facts collection techniques.
- iv.) Record the facts using such documents like; grid or x chart, document analysis form output, File analysis form, output analysis form, clerical procedure flowchart, procedure narrative.
- v.) Analyse.

4. System Design and Development:-The next step is design and development of any new systems or applications that are required to satisfy the need. The systems specification must describe how the new system will work. Screen layouts and report formats or outputs must be designed, hardware, software, people, network, including the input data needed, and the general processing procedures required to convert data into information products that will satisfy the functional requirements of the proposed system by means of program specifications, structure charts, pseudo code or flowcharts. Then, the programmers must code, test and debug all the programs in the systems.

5. System implementation:-This is the phase of the system development when the new system become operational . It is acquisition and integration of the physical and conceptual resources that produce a working system. That is, the stage when people actually begin to use a new system.

6. Review and Maintenance: - The use this process is to monitor, evaluate, and modify the system as needed. Maintenance involves keeping the system up to the date and the system could be subjected to review to ensure that it conforms to the requirements identified in the feasibility study in order to get an improved system.

CHANGEOVER TO THE NEW SYSTEM

Conversion procedure concludes the development and implementation of the new system. After the conversion process, the system is put to use.

1.) Direct changeover is a sudden changeover method. It involves discarding with the old system and setting into operation of the new system.

Features of direct changeover are as follows:

i.) There must be assurance in the new system's reliability before it can be employed.

ii.) It is used when the two system are totally different or incompatible with each other.

2.) Parallel changeover involves running both the old and the system concurrently, using the same data. This method is characterized by the following:

i.) The method requires employing extra staff to cope with possible increase in the work load.

ii.) It provides management with an opportunity and facility of testing of the new system while the old system is still in operation.

3.) Phase changeover, the new system is put into use one part at a time. The conversion can be phased in modules of the system or phased running the whole module department by department for or different geographical areas. Phase conversion is very popular for large-scale system.

4.) Pilot conversion is a trial system implemented in a subset of the overall operation. If It is successful then it is brought into other areas.

SYSTEM TESTING

Often it is not immediately obvious that there are logical errors in a program, indeed some programs may function correctly for months or even years before a particular circumstance, or piece of input data, causes a problem or error to surface.

It is responsibility of a programmer to test, as far as possible, that all parts of the program work correctly.

Types of testing:

a.) Functional testing: the system is tested to make sure all parts of it perform correctly using test data.

b.) Recovery testing: The purpose of recovery testing is to ensure that the system can recover from various types of failure. Recovery testing may involve simulating hardware failures and power failures.

c.) Performance testing: The purpose of this kind of testing is to ensure that the system can handle the volume of data that it will have to cope with, under the conditions that will prevail in the user environment

SYSTEM MODELS

A model is an abstraction of something known in real life situation. It represents real world element or group of elements (some objects or activity) which is called entity. Managers use models to represent problems to be solved. The objects or activities that cause problems are the entities.

Types of models.

There are four basic types of models, namely;

i.) Physical models, (ii.) Narrative models, (iii.) Graphic models, and (iv.) Mathematical models.

1. Physical model: A physical model is a 3-dimensional representation of its entity. Physical models used in the business world include scale models of shopping centre and prototypes of new automobiles.

2. Narrative Models: Narrative model describes its entity with spoken or written words, where both the listener and reader can understand the entity from the narrative. It is a type of model that managers use daily.

3. Graphic Model: A graphic model represents its entity with an abstraction of lines, symbols, or shapes. This is another type model in constant use in business to communicate information to managers.

4. Mathematical Model

Any mathematical formula or equation is a mathematical model. The mathematical model accounts for most of the current interest in business modeling.

Uses of Models

i. Facilitate understanding.

ii. Facilitates communication.

iii. Predict the future.